Neurocomputing 672 (2026) 132770

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

Knowledge concept cold-start approach for cognitive diagnosis

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ARTICLE INFO

Communicated by R. Mao

Keywords:
Cognitive diagnosis
Neural network
Cold-start

ABSTRACT

Domain-level zero-shot cognitive diagnosis aims to assess students in the absence of interaction records. However, existing methods in this area primarily analyze student-exercise interactions while neglecting the cold-start problem at the knowledge concept level. For newly introduced knowledge concepts in a course, interaction records are missing, making it challenging for the system to accurately infer students' proficiency based on historical data. To address this issue, we propose the Knowledge Concept Cold-Start Cognitive Diagnosis Model (KCSCD). KCSCD comprehensively considers both interaction and interaction missing knowledge concepts. It introduces a Knowledge Concept Domain Inference Module to model students' deep knowledge states and a Deep Learning Ability Assessment Module to evaluate their deep learning ability. These components are then integrated through a gating mechanism for weighted fusion. Experiments conducted on the ASSIST17, ASSIST09, and Junyi datasets demonstrate that KCSCD outperforms existing methods in both accuracy and interpretability. Moreover, the model exhibits a significant advantage in predicting students' knowledge mastery in cold-start scenarios. Our code is available at https://github.com/lihihuan1213/KCSCD.

1. Introduction

Cognitive Diagnosis Modeling (CDM) is a crucial foundational work in intelligent education systems [1–3]. It enables educators to precisely assess students' mastery of specific knowledge concepts by analyzing their interactions with practice questions throughout the learning process [4]. Traditional educational assessment methods primarily rely on exam scores or standardized tests, which often fail to pinpoint students' weaknesses in particular knowledge areas. CDM addresses this limitation by constructing detailed models that differentiate students' proficiency levels across various knowledge concepts within a subject. The insights derived from cognitive diagnosis modeling can further support downstream educational applications, such as knowledge tracing [5–8] or customized study path suggestions [9–11], ultimately enhancing learning efficiency through tailored instruction.

Over the years, researchers have made substantial efforts in cognitive diagnosis work and have achieved remarkable results [12,13]. The effectiveness of Cognitive Diagnosis Modeling (CDM) largely relies on interaction functions that capture the complexity of student-exercise interaction. Initial studies mainly used manual designs to model interaction functions [14,15]. Later, these functions were optimized through the integration of Deep Neural Networks (DNNs) [16–18] and Graph Neural Networks (GNNs) [19–23], leading to significant improvements in model performance. Current deep interaction models mainly follow traditional Item Response Theory (IRT) [14], which estimates the probability of correct answers using students' ability levels and question difficulty. Existing cognitive diagnosis models typically assess students' mastery of knowledge concepts through their correctness in answering specific questions [1,9,24,25]. CDMs have limitations when applied to students with sparse interaction logs. In cognitive diagnosis models, the interaction logs between students and tasks serve as crucial evidence for evaluating students' cognitive states. To address this issue, the Domain-level Zero-shot Cognitive Diagnosis (DZCD) task was introduced. DZCD aims to perform effective cognitive diagnosis for unseen students in a new domain without interaction logs. By leveraging DZCD, researchers hope to overcome the limitations caused by sparse interaction logs, enabling cognitive diagnosis models to offer more accurate personalized assessments in various educational settings, ultimately supporting students' learning and development.

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https://doi.org/10.1016/j.neucom.2026.132770
Received 11 July 2025; Received in revised form 5 December 2025; Accepted 17 January 2026
Available online 22 January 2026
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Lately, researchers have made significant progress in addressing the DZCD task, with promising solutions such as TechCD [26] and the Zero-1-to-3 framework [24]. Existing methods typically prioritize the student-exercise interaction as the main consideration, overlooking the impact at the knowledge concept level. For instance, in intelligent education systems, when a new student first uses an online learning platform, the system struggles to accurately assess their knowledge level due to the lack of historical interaction data. This limits the effectiveness of personalized learning recommendations and learning path planning. Similarly, when new knowledge concepts are introduced into a course, existing cognitive diagnosis models often struggle to directly infer students' mastery of these new concepts. For these newly introduced concepts, students' exercise records remain blank, and the system cannot utilize existing data for effective diagnosis and prediction. This leads to a significant reduction in diagnostic accuracy when cognitive diagnosis models face new knowledge concepts, particularly in the early stages when there is insufficient interaction data between students and these concepts. We refer to these newly introduced concepts as Interaction Missing Knowledge Concepts (MKCs), while the knowledge concepts covered in students' previous practice responses are termed Interaction Knowledge Concepts (IKCs). In real-world educational settings, the limited number of exercises means that each student typically interacts with only a subset of knowledge concepts during tests or assignments, further complicating the knowledge concept cold-start problem. As a result, relying solely on these limited assessment items prevents educational systems from comprehensively evaluating students' knowledge states and learning abilities. This limitation affects both the accuracy of student evaluations and the effectiveness of personalized learning approaches. By analyzing the ASSIST17 dataset [27], we find that most students engage with only a few course concepts. As shown in Fig. 1, approximately 15.7% of students interact with a relatively large number of knowledge concepts, while most students cover only half of the total knowledge concepts in the course. This phenomenon reveals an imbalance in students' cognitive states and highlights the commonality and severity of the knowledge concept cold-start problem. This limitation is not only evident in the ASSIST17 dataset but has also been observed in many other open datasets. Therefore, effectively addressing the cold start problem of knowledge concepts remains a key area for improvement in cognitive diagnosis. When new knowledge concepts are introduced, students may not have sufficient practice to assess and grasp these concepts, and cold starts can help address this issue. By introducing cold starts at the Knowledge Concept (KC) level, assessments can be conducted as soon as students are first exposed to new knowledge, thereby identifying and correcting mistakes in a timely manner and improving learning efficiency. Cold starts can provide personalized learning paths for each student, adjusting according to their initial capabilities and needs, which helps evaluate the students' initial abilities when they encounter new knowledge concepts, providing a reference for subsequent teaching. Teachers can adjust teaching strategies based on the results of cold starts, offering personalized guidance and support for different students.

In order to solve the problem of interaction missing knowledge concepts, we propose the Knowledge Concept Cold-Start Cognitive Diagnosis (KCSCD) model. Unlike previous studies, we approach the problem of interaction missing knowledge concepts from a new perspective by defining it as a cold-start cognitive diagnosis problem within knowledge concept domains. To diagnose students' performance on exercises containing interaction missing knowledge concepts, an effective method must establish connections between Interaction Knowledge Concepts (IKCs) and Missing Knowledge Concepts (MKCs). The KCSCD model consists of three main modules: (1) Knowledge concept Domain Inference Module based on the scoring mechanism. This module is based on a scoring mechanism. It calculates similarity scores between knowledge concepts to construct a similarity score matrix. By integrating the student's current knowledge mastery level, it infers their deep knowledge state, which includes MKCs. In educational systems, knowledge concept embedding vectors are commonly used to represent students' learning content or skills [28], such as "trigonometry", "calculus", etc. These concepts are typically associated with specific knowledge systems or curriculum structures [29]. We define a vector space for the knowledge concepts and map each concept to this space, and measure the similarity between different knowledge concepts by calculating the vector distance between them. (2) Deep Learning Ability Assessment Module. This module assesses deeper learning abilities by combining the initial embedding vectors of students' learning abilities with the feature embedding vectors of exercises. Students' learning abilities are a critical factor in cognitive diagnosis tasks, as they directly influence personalized education and the effectiveness of teaching and learning. Students with higher learning abilities tend to have higher accuracy rates than those with lower abilities when encountering exercises involving MKCs. Educational systems or instructors can tailor exercises to students' individual learning abilities, preventing frustration or boredom caused by exercises that are either too challenging or overly simple, thereby enhancing learning efficiency. (3) Hybrid Latent Trait Prediction Module. This module integrates students' deep knowledge states and learning abilities through a gating mechanism. Inspired by ACD [2], we incorporate students' emotional factors into the model, modeling these emotional factors as guessing and slipping. Following the DINA model [30], we generate the final prediction based on these factors.

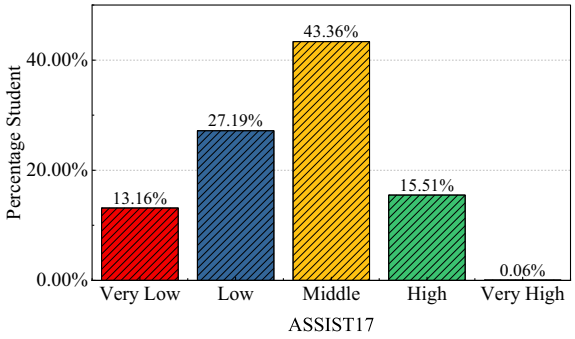


Fig. 1. The proportion of the number of students interacting with different levels of concepts from the ASSIST17 data set.

The technical contributions of our work can be summarized as follows:

- We propose a novel cognitive diagnosis model, KCSCD, which takes into account both IKCs and MKCs and improves prediction accuracy in cold-start scenarios of knowledge concepts.
- We have designed a knowledge concept domain inference module and a deep learning ability assessment module to enhance the ability to diagnose MKCs and accurately assess students' learning abilities.
- Through experiments on three benchmark datasets, we demonstrate that our KCSCD model surpasses existing methods in effectiveness and generalizability.

2. Related works

Here we present three closely related research areas: (1) Cognitive Diagnosis Modeling, (2) Cold-Start Cognitive Diagnosis, and (3) Interaction Missing Knowledge Concepts Research.

2.1. Cognitive diagnosis modeling

In intelligent educational systems, Cognitive Diagnosis Modeling (CDM) is a core task that analyzes students' test performance to infer their mastery of specific knowledge concepts [31]. Current cognitive diagnosis methods predominantly rely on classical paradigms such as Item Response Theory (IRT) [14] and Deterministic Input, Noisy "And" gate (DINA) [30]. IRT serves as a foundational model in cognitive diagnosis,

using manually designed functions to describe the connection between student and item characteristics. This theory assumes that student performance is jointly influenced by latent ability and item parameters. To enhance expressiveness, the Multidimensional Item Response Theory (MIRT) [15] extends the single-dimensional features into a multidimensional vector. This extension enables multiple dimensions of student ability to concurrently influence test performance, making it well-suited for handling multidisciplinary or complex knowledge structures. Neural Cognitive Diagnosis (NCD) [17] improves upon traditional IRT models by replacing manually designed functions with neural networks, which automatically extract and learn higher-order features between student performance and item characteristics. This approach enhances prediction accuracy and is more effective for processing large-scale datasets. RCD [20] integrates multiple local graphs containing student, exercise, and concept relationships through a multi-level attention network, balancing the aggregation across different graphs. This approach integrates information from multiple perspectives, enhancing the understanding and evaluation of students' cognitive states. Song et al. propose a CMNCD [9], which leverages deep neural networks to integrate the cross-modal content information of exercises, thereby modeling students' performance more accurately. Affect-aware Cognitive Diagnosis (ACD) [2] further enhances cognitive diagnosis by incorporating students' emotional state modeling. It considers both cognitive and affective states, such as focus and confusion, which significantly impact learning performance. By modeling emotional states, ACD offers more nuanced assessments of student ability and provides personalized learning recommendations. Symbolic Cognitive Diagnosis (SCDM) [13] aims to enhance the interpretability of cognitive diagnosis and maintain good generalization performance by combining symbolic tree and gradient optimization methods. It effectively combines discrete symbolic representation with continuous parameter optimization by alternately optimizing symbolic representation and parameters.

Compared with previous studies, this research further expands the model's application scenarios based on the basic modeling method of guessing factor and slip factor in traditional cognitive diagnosis models. We focus on addressing a critical issue in real-world educational scenarios, namely the absence of students' historical interaction data, and introduce Domain-level Zero-shot Cognitive Diagnosis (DZCD). To enhance the model's diagnostic accuracy for knowledge concepts without interaction records, we have specifically designed the Knowledge Concept Domain Inference Module and the Deep Learning Ability Assessment Module, aiming to achieve comprehensive and accurate inference of students' knowledge states.

2.2. Cold-start cognitive diagnosis

The cold-start problem is a key challenge in machine learning and recommendation systems [32,33]. When new courses or concepts enter the system, the lack of historical data prevents accurate evaluation of preferences or similarities, making recommendations difficult. In intelligent education, the cognitive diagnosis cold-start problem arises when new students or exercises lack sufficient learning history, hindering accurate assessment of cognitive states. This issue has two main types: student cold-start and exercise cold-start. Student cold-start occurs when a cognitive diagnosis system encounters a new student [34]. Without prior data on their knowledge state, the system struggles to assess their proficiency and learning needs. Exercise cold-start happens when new exercises or learning materials are introduced [35]. Due to limited interaction records, the system cannot accurately determine exercise difficulty or its effectiveness in distinguishing students' knowledge levels. Recent studies address these challenges using graph-based and transfer learning techniques. The TechCD framework [26] leverages a teaching knowledge concept graph to facilitate the transfer of characteristics and data from students in existing domains to the target domain, thereby enhancing the model's ability for zero-shot diagnosis. Gao et al. [24] proposed a new strategy that leverages the behavior patterns of early bird students to simulate the practice logs of cold-start students. Early bird students are those who are active on the platform and have a substantial amount of learning behavior, and the system can analyze their behavior patterns to infer the learning performance of cold-start students in the early stages.

The TechCD framework employs a Graph Convolutional Network (GCN) [36] on pedagogical knowledge graphs to learn transferable student cognitive states and domain-specific exercise features. This allows student cognitive signals to transfer from established domains to cold-start scenarios.

This paper introduces a new perspective: knowledge concept cold-start. When new knowledge concepts are introduced, they often remain untested in assessments, making it difficult to evaluate students' mastery of these concepts. The primary challenge is linking tested (interaction) and untested (interaction-missing) knowledge concepts. To address this, we construct a vector space of knowledge concepts and compute their similarities, enabling the connection between known and unknown concepts for more comprehensive cognitive diagnosis, which ensures that KCSCD maintains high prediction accuracy while possessing excellent low-latency inference capabilities, flexible structural scalability, and superior online adaptability. The model is especially suitable for intelligent teaching platforms that require frequent updates of student states and dynamic introduction of new knowledge concepts, providing technical support and application potential for practical deployment.

2.3. Interaction missing knowledge concepts research

Cognitive diagnosis of interaction-missing knowledge concepts is a key challenge in smart education, aiming to infer students' mastery of concepts they have not directly interacted with. Recent advancements leverage deep neural networks and contextual features to enhance student and exercise representations [20,37,38]. The Knowledge-sensed Cognitive Diagnosis (KSCD) framework [39] addresses this limitation by learning these intrinsic relationships from students' response logs to infer their overall knowledge mastery. This method can interpret students' understanding of interaction missing knowledge concepts and, by integrating the student-exercise interaction model, provides more precise predictions of students' knowledge levels. These studies provide deeper insights into students' knowledge states, supporting improved teaching strategies and personalized learning experiences. Future research can further refine the construction and utilization of knowledge relationships to enhance cognitive diagnosis accuracy.

3. Problem definition

In an intelligent education system, we consider a set of N students, M exercises, and K knowledge concepts, which are defined as follows: Student set $S = \{s_1, s_2, \ldots, s_N\}$, Exercise set $E = \{e_1, e_2, \ldots, e_M\}$, and knowledge concept set $C = \{c_1, c_2, \ldots, c_K\}$. The relationship between knowledge concepts and exercises is captured using a Q-matrix, denoted as $\boldsymbol{Q} = \{Q_{jk}\}_{M\times K} \in \{0,1\}_{M\times K}$. Here, if exercise e_j is related to concept c_k, then $Q_{jk} = 1$; otherwise, $Q_{jk} = 0$. We represent the interaction log between student s and exercises as a set of triplets (s, e, r), where $r \in \{0, 1\}$ represents the score a student s obtained on exercise e.

Our objective is to learn an interaction function $\mathcal{F}_\theta(\cdot)$ that captures the complex interaction patterns between students, exercises, and knowledge concepts. It utilizes students' performance in different exercises recorded in learning logs, combined with the Q-matrix, to assess their mastery of knowledge concepts. This approach remains effective even in cases of missing interactions, enabling accurate evaluation of students' mastery levels of knowledge concepts. Such capability not only supports personalized learning interventions but also enhances the adaptability of educational systems, making them more intelligent and efficient.

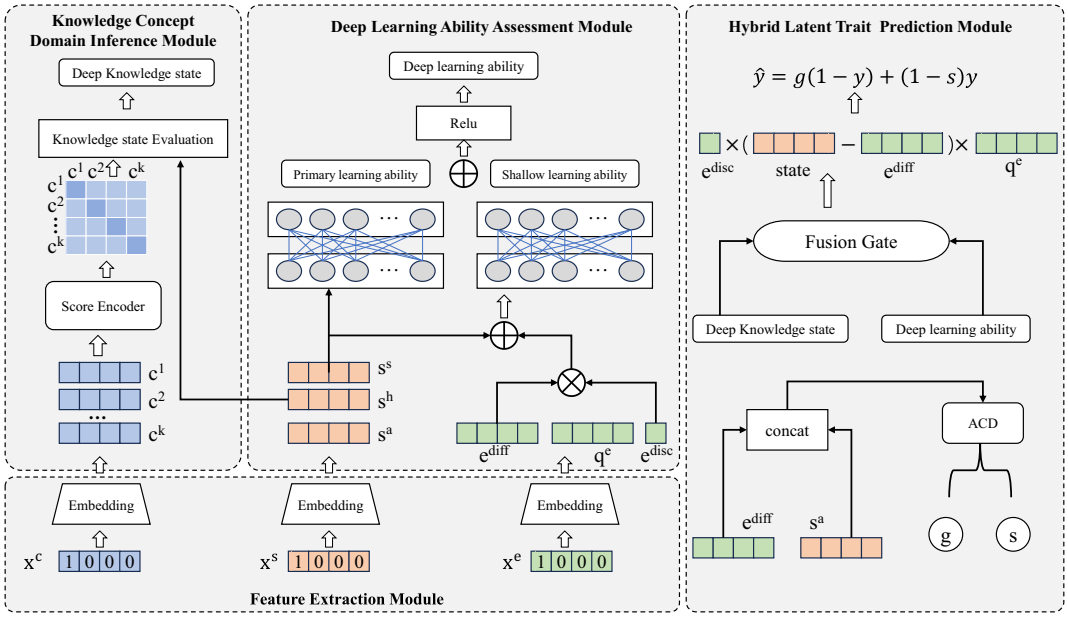


Fig. 2. The overview architecture of KCSCD.

4. Knowledge concept cold-start cognitive diagnosis

In this section, we present a comprehensive overview of the KCSCD model, as shown in Fig. 2. The model is composed of four main components: (1) the feature extraction module, (2) the knowledge concept domain inference module, (3) the deep learning capability assessment module, and (4) the hybrid latent trait prediction module.

4.1. Feature extraction module

4.1.1. Student factors

In order to analyze the students more comprehensively, we design three vectors s_i^h, s_i^s and s_i^a for each student. s_i^h represents the extent to which the i-th student has mastered each of the knowledge concepts. s_i^s denotes the learning ability vector of the i-th student. s_i^a denotes the latent affective trait of the i-th student. It is worth noting that s_i^a comes from ACD. Assuming x^s is the one-hot vector of the student, and $\mathbf{S}_h$, $\mathbf{S}_s$, and $\mathbf{S}_a$ are trainable matrices, we can obtain s_i^h, s_i^s, and s_i^a by multiplying $\boldsymbol{x}^s$ with them respectively:

$$s_i^h = \boldsymbol{x}^s \times \mathbf{S_h}, \tag{1}$$

$$s_i^s = \boldsymbol{x}^s \times \mathbf{S_s}, \tag{2}$$

$$s_i^a = \boldsymbol{x}^s \times \mathbf{S_a}, \tag{3}$$

where $s_i^h \in (0,1)^{1\times K}$, $s_i^s \in (0,1)^{1\times K}$, $s_i^a \in (0,1)^{1\times d_1}$, $\boldsymbol{x}^s \in \{0,1\}^{1\times N}$, $\mathbf{S}_h \in \mathbb{R}^{N\times K}$, $\mathbf{S}_s \in \mathbb{R}^{N\times K}$, $\mathbf{S}_a \in \mathbb{R}^{N\times d_1}$. The parameter d_1 indicates the dimension of latent affective traits.

4.1.2. Exercise factors

Assuming that $\boldsymbol{x}^e$ is the one-hot vector of the exercise and $\mathbf{E}_1$, $\mathbf{E}_2$ are the trainable matrices, through calculation we can obtain the knowledge-related question vector $\boldsymbol{q}_j^e$, the question discrimination scalar e_j^{disc} and the question difficulty vector $\boldsymbol{e}_j^{diff}$ which contains the knowledge concept information in the questions. The computational procedure is as follows:

$$\boldsymbol{q}_j^e = \boldsymbol{x}^e \times \mathbf{Q}, \tag{4}$$

$$e_j^{disc} = \boldsymbol{x}^e \times \mathbf{E}_1, \tag{5}$$

$$\boldsymbol{e}_j^{diff} = \boldsymbol{x}^e \times \mathbf{E}_2, \tag{6}$$

where $\boldsymbol{q}_j^e \in (0,1)^{1\times K}$, $e_j^{disc} \in (0,1)$, $\boldsymbol{e}_j^{diff} \in (0,1)^{1\times K}$, $\boldsymbol{x}^e \in \{0,1\}^{1\times M}$, $\mathbf{E}_1 \in R^{M\times 1}$, $\mathbf{E}_2 \in R^{M\times K}$, and $\mathbf{Q} \in \{0,1\}^{M\times K}$ indicates which knowledge concepts are included in the question $\boldsymbol{e}_j$.

4.1.3. Knowledge concept factors

Assuming $\boldsymbol{x}^c$ is the one-hot vector of a knowledge concept and $\mathbf{C}$ is a trainable matrix, we obtain the knowledge concept vector $\boldsymbol{c}_k$ through the following computational procedure:

$$\boldsymbol{c}_k = \boldsymbol{x}^c \times \mathbf{C}, \tag{7}$$

where $\boldsymbol{c}_k \in (0,1)^{1\times d_2}$, $\boldsymbol{x}^c \in \{0,1\}^{1\times K}$, $\mathbf{C} \in (0,1)^{K\times d_2}$, and d_2 denotes the dimension of the knowledge concept vector.

4.2. Knowledge concept domain inference module

In the knowledge concept cold-start scenario, students lack interaction logs with newly introduced knowledge concepts, i.e., MKC. Traditional methods struggle to provide comprehensive and accurate assessments, which impacts the implementation of downstream educational strategies. This limitation affects the effectiveness of diagnosis and personalized teaching design. Therefore, we propose the Knowledge Concept Domain Inference Module (KCDIM). The core of KCDIM is a scoring mechanism that analyzes and understands the similarity relationships between knowledge concepts. Specifically, we define a knowledge concept vector space, where we use student response logs to learn a student-knowledge state matrix $\mathbf{P} \in \mathbb{R}^{N\times K}$. Therefore, one row of the matrix P represents a student's mastery level of each knowledge concept s_i^h. Typically, for any two knowledge concepts, the more similar they are to each other, the more similar students' mastery of them should be. Based on this assumption, we use cosine similarity to calculate the distance between each knowledge concept pair and use the result as a measure of knowledge concept similarity. After the above calculations, we obtain the knowledge concept similarity score matrix $\mathbf{M}^{\omega} \in \mathbb{R}^{K\times K}$, which does not depend on the students' mastery vectors; instead, it is constructed independently of the mastery status, based on structural and semantic adjacency. This design effectively avoids the self-reinforcing assumption of "predicting mastery with mastery" and ensures stable similarity estimation capability under cold-start conditions. By combining these scores with the student's proficiency in IKCs, we can infer the student's proficiency in MKCs. Even if students have

not directly encountered certain knowledge concepts, we can still infer their potential knowledge states by analyzing the concepts they have already mastered. The knowledge concept similarity matrix is calculated as follows:

$$\mathbf{M}^{\omega} = Score(\mathbf{C}), \tag{8}$$

where $\mathbf{M}^{\omega} \in (0,1)^{K\times K}$ is the knowledge concept similarity score matrix. $Score(\cdot)$ represents the process of evaluating knowledge concepts to obtain similarity scores between concept pairs. These scores not only reflect the surface similarity between knowledge concepts but also capture deeper semantic and logical connections. We define the set of knowledge concepts corresponding to the obtained similarity scores as the domain where the MKCs c_k belongs. The similarity scores of each concept form the similarity score matrix $\mathbf{M}^{\omega}$. Finally, we multiply the obtained similarity score matrix $\mathbf{M}^{\omega}$ by the student's knowledge state vector $\boldsymbol{s}_i^h$ to obtain a student's deep knowledge state vector $\tilde{\boldsymbol{s}}_i^h \in (0,1)^{1\times K}$. This vector reflects their potential mastery of both IKCs and MKCs. The formula is as follows:

$$\tilde{\boldsymbol{s}}_i^h = \boldsymbol{s}_i^h \times (\mathbf{M}^{\omega})^T, \tag{9}$$

where, $\tilde{\boldsymbol{s}}_i^h$ represents the deep student knowledge state vector, which indicates the mastery level of the i-th student over all knowledge concepts. It includes both interaction knowledge concepts and missing knowledge concepts.

4.3. Deep learning ability assessment module

Students exhibit individual differences in learning ability, which are reflected not only in their mastery of interactive knowledge concepts but also in their responses to interaction-missing knowledge concepts. When new knowledge concepts are introduced into a course, students who have a strong capacity for learning tend to grasp them more quickly, while those with weaker learning abilities may face greater challenges. In real-life scenarios, students with higher learning abilities tend to solve more difficult problems with greater accuracy compared to those with lower learning abilities. In the knowledge concept cold-start scenario, we lack interaction logs between students and interaction-missing knowledge concepts. Therefore, assessing students' learning abilities becomes crucial for cognitive diagnosis tasks, as it enables the model to more accurately predict a student's ability to correctly answer exercises related to interaction-missing knowledge concepts. To evaluate students' deep learning abilities, we have designed a learning ability assessment module. After obtaining the embedding factors of students and exercises, to characterize students' potential adaptability to unknown exercises in knowledge cold-start scenarios, we introduce two exercise attribute vectors: the exercise difficulty vector and the exercise discrimination vector. The exercise difficulty reflects the cognitive challenge level of the question, while discrimination measures the sensitivity of the exercise in distinguishing students of different ability levels. This result is then combined with the student's learning ability embedding and passed through a sigmoid function to map it to a value between (0, 1). This mapped value is passed through a network layer to compute the student's current learning ability. To prevent the loss of important information during the learning process, we apply residual linkage. The calculation process of students' deep learning ability $\tilde{\boldsymbol{s}}_i^s \in (0,1)^{1\times K}$ is as follows:

$$\boldsymbol{s}_i^0 = F_1\left(\sigma\left(\boldsymbol{s}_i^s + \boldsymbol{e}_j^{disc} \times \boldsymbol{e}_j^{diff}\right)\right), \tag{10}$$

$$\tilde{\boldsymbol{s}}_i^s = \mathrm{ReLU}\left(\boldsymbol{s}_i^0 + F_2(\boldsymbol{s}_i^s)\right), \tag{11}$$

where F_1 and F_2 denote fully connected layers, while $\sigma(\cdot)$ and Relu$(\cdot)$ represent the sigmoid and Relu activation functions, respectively.

4.4. Hybrid latent trait prediction module

In a knowledge concept cold start scenario, whether or not a student gets a question right is influenced by internal factors from the student themselves as well as external factors from the topic and concept. The internal factors of students include their learning ability, mastery of knowledge concepts, as well as guessing and slipping behaviors. External factors involve the exercise's difficulty, discrimination and concepts. Therefore, in our prediction module, we take both internal and external factors. The prediction module takes as input the student's knowledge state vector $\tilde{\boldsymbol{s}}_i^h$, learning ability vector $\tilde{\boldsymbol{s}}_i^s$, exercise embedding vector $\boldsymbol{e}_j$, exercise difficulty parameter $\boldsymbol{e}_j^{diff}$, and the knowledge concept vector involved in the exercise $\boldsymbol{q}_j^e$, and outputs the predicted response y. To better integrate the student's deep knowledge state and deep learning ability, we introduce a gating mechanism and use $\tilde{\boldsymbol{s}}_i^h$ and $\tilde{\boldsymbol{s}}_i^s$ as inputs:

$$g_{gate} = G(\tilde{\boldsymbol{s}}_i^h, \tilde{\boldsymbol{s}}_i^s), \tag{12}$$

$$\boldsymbol{s}_{gate} = g_{gate} \times \tilde{\boldsymbol{s}}_i^h + (1 - g_{gate})\tilde{\boldsymbol{s}}_i^s, \tag{13}$$

where $G(\cdot)$ denotes the gating mechanism, $\boldsymbol{s}_{gate} \in (0,1)^{1\times K}$ denotes the student's final mastery of the knowledge concept. Then, we can obtain the initial prediction as follows:

$$y = \boldsymbol{e}_j^{disc} \times (\boldsymbol{s}_{gate} - \boldsymbol{e}_j^{diff}) \times \boldsymbol{q}_j^e. \tag{14}$$

Inspired by ACD [2], we introduce the student's latent affective trait vector $\boldsymbol{s}_i^a$ into this module. We use $\boldsymbol{s}_i^a$ to model the student's guessing and slipping probabilities. Here, g represents the probability of answering an exercise correctly due to guessing, while s represents the probability of making an error due to slipping. Specifically, the affective perception module computes the guessing and slipping probabilities as follows:

$$a = \sigma(W_a \times [\boldsymbol{s}_i^a, \boldsymbol{e}_j^{diff}] + b_a), \tag{15}$$

$$g = \sigma(W_g \times a + b_g), \tag{16}$$

$$s = \sigma(W_s \times a + b_s), \tag{17}$$

where $W_g \in (0,1)^{1\times d}$ and $W_s \in (0,1)^{1\times d}$ are mapping matrices, while b_g and b_s are bias terms, $[\cdot,\cdot]$denote the concatenation operation. Finally, the model predicted response is:

$$\hat{y} = g(1-y) + (1-s)y. \tag{18}$$

This multi-factor fusion method integrates the student's mastery level, learning ability, and affective traits, leading to more accurate response predictions for each exercise. At the same time, a thorough analysis of students' knowledge and affective traits allows for better identification of their needs, forming a strong foundation for personalized teaching and learning interventions. This approach provides targeted educational support, helping students learn new concepts more effectively. Given the model's predicted value $\hat{y}$ and the student's actual response $r \in \{0,1\}$, where $r = 1$ indicates that the student answered the question correctly and $r = 0$ indicates that the student answered incorrectly, we use the binary cross-entropy loss function as the loss metric for the model:

$$\mathcal{L} = -\sum_i \left(r_i \log \hat{y}_i + (1 - r_i)\log(1 - \hat{y}_i)\right). \tag{19}$$

Not all datasets contain affective labels. Inspired by ACD, for datasets without affective labels, we adopt an alternative strategy to obtain the affective loss $\mathcal{L}_a$. Positive emotions are typically associated with correct answers, while negative emotions are often linked to incorrect responses. Therefore, we classify students whose answers match those of the anchor student as positive samples and those whose answers differ as negative samples. Affective loss is then obtained through contrastive learning. The predicted loss of the model is as follows:

$$\mathcal{L}_a = -\sum_i \frac{1}{|\mathcal{T}_i|} \sum_{j\in\mathcal{T}_i} log \frac{\phi(\hat{\boldsymbol{s}}_i^a, \hat{\boldsymbol{s}}_j^a)}{\sum_{k\in\mathcal{F}_i} \phi(\hat{\boldsymbol{s}}_i^a, \hat{\boldsymbol{s}}_k^a)}, \tag{20}$$

where, $\phi(\hat{\boldsymbol{s}}_i^a, \hat{\boldsymbol{s}}_j^a) = \exp[\mathrm{sim}(\hat{\boldsymbol{s}}_i^a, \hat{\boldsymbol{s}}_j^a)/\tau]$, $\mathrm{sim}(\cdot,\cdot)$ represents the cosine similarity between two estimated affective vectors, and τ is the temperature

Table 1
Statistics of experimental datasets.

Dateset	ASSIST17	ASSIST09	Junyi
Students	1709	4163	10,000
Exercises	3162	13,016	835
Knowledge concepts	102	123	835
Response records	311,569	267,867	220,799

parameter. $\mathcal{T}_i$ denotes the set of positive pairs of students in the i-th exercise, while $\mathcal{F}_i$ denotes the set of negative pairs. By combining the model's prediction loss $\mathcal{L}$ and the affective loss $\mathcal{L}_a$, we obtain the total loss:

$$\mathcal{L}_{KCSCD} = \mathcal{L} + \lambda \mathcal{L}_a, \tag{21}$$

where λ represents the trade-off hyperparameter that balances the two loss terms.

5. Experiment

We perform a number of experiments on three real-world datasets and analyze the results. The key questions we aim to address through these experiments are:

RQ1: How does KCSCD perform overall compared to other baseline models?

RQ2: How do students' deep knowledge states and deep learning abilities affect the model's performance?

RQ3: How do hyperparameters affect KCSCD?

RQ4: Can KCSCD effectively solve the knowledge concept cold-start problem?

RQ5: How does the interpretability of KCSCD perform?

5.1. Experimental settings

5.1.1. Dataset description

The experiments were conducted on three real-world educational datasets: ASSIST17 [27], ASSIST09 [40], and Junyi [41]. The statistical information of these datasets can be found in Table 1, and their detailed descriptions are as follows:

- **ASSIST17**[1]: The ASSIST2017 dataset comes from the ASSISTments platform and records student behavior data during their math learning process. It provides information such as students' answering history, correctness, learning progress, and emotional data, such as concentration and boredom, during the answering process. This makes it uniquely valuable for researching the impact of student emotions on learning outcomes.
- **ASSIST09**[2]: The ASSIST2009 dataset also comes from the ASSISTments platform but only includes student learning behavior data without considering emotional factors. This dataset records students' math learning data from the 2009–2010 academic year, mainly covering students' answering performance, learning progress, and skill mastery.
- **Junyi**[3]: The Junyi dataset comes from an online education platform and records the online learning behavior data of a large number of students across various subjects. This dataset includes records of student interactions with the platform, covering learning time, answer correctness, and various activities students participated in on the platform, but it does not include emotional factor analysis.

To ensure the quality of model training and evaluation, we perform strict data preprocessing and filtering. First, we exclude students with fewer than 15 learning records, ensuring a sufficiently rich learning history to avoid bias from data sparsity. Meanwhile, the training set, validation set, and test set are divided in an 8:1:1 ratio. These three datasets are publicly available and widely used in the field of cognitive diagnosis. They provide rich student learning behavior data, offering standardized benchmarks for model experimentation and validation, making it easier to compare with other research. Therefore, these datasets are the ideal choice for our experiments.

5.1.2. Baselines

In order to verify the validity of KCSCD, we will compare the performance with seven baselines:

- **DINA [30]:** The Deterministic Input, Noisy-And Gate (DINA) model is a simple cognitive diagnosis model that estimates students' mastery of knowledge concepts by incorporating guessing and slipping parameters.
- **IRT [14]:** The Item Response Theory (IRT) model combines students' latent abilities with the characteristics of the items to predict the probability of a student answering each question correctly.
- **MIRT [15]:** The Multidimensional Item Response Theory (MIRT) model is an extension of the IRT model that allows for modeling students' abilities and item characteristics in a multidimensional space.
- **NCD [17]:** The Neural Cognitive Diagnosis (NCD) model uses neural networks to automatically learn the complex interactions between students and items, improving the accuracy and robustness of cognitive diagnosis.
- **RCD [20]:** The RCD model constructs a knowledge concept relationship graph and uses graph neural networks to capture the dependencies between knowledge concepts.
- **KSCD [39]:** The KSCD framework aims to learn the intrinsic relationships among concepts from student response logs, and integrate these relationships to infer students' mastery of all concepts.
- **Zero-1-to-3 [24]:** The framework known as Zero-1-to-3 aims to achieve domain-level zero-shot cognitive diagnosis through a batch of data from an early group of students, and is optimized towards three diagnostic objectives.
- **SCDM [13]:** The Symbolic Cognitive Diagnosis Model (SCDM) combines symbolic trees to explicitly represent the complex student-exercise interaction function and uses gradient-based optimization methods to efficiently learn student and exercise parameters.
- **ACD [2]:** The Affective Cognitive Diagnosis (ACD) model incorporates students' emotional states, modeling their affective traits during the answering process to improve the accuracy of cognitive diagnosis.

5.1.3. Parameter setting

The experiments were conducted on an RTX 4090 server. We used the Adam optimizer for training [42]. To ensure training stability and convergence, we initialized model parameters using Xavier initialization. The weights are initialized with Xavier initialization, and the biases are set to 0. Hyperparameters in the joint loss function were adjusted to balance different losses, with the parameter λ set to 1 and the knowledge concept embedding dimension set to 256. The learning rate is 0.001 and the batch size is set to 128. Moreover, when students generate new interaction data, the system only needs to update their embedding representations without retraining the entire model. When new knowledge concepts are introduced in the courses, it is only necessary to generate vector representations of the new concepts in the embedding space and calculate their similarity scores with existing concepts, integrating them quickly into the original knowledge concept similarity matrix. This process does not require retraining the main model parameters, significantly enhancing the system's adaptability to new knowledge.

[1] https://sites.google.com/view/assistmentsdatamining/dataset
[2] https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data
[3] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=1198

Table 2
Comparison of KCSCD and baseline models on three publicly available datasets. Best results in bold, next best underlined. Improvement Rate denotes the relative performance improvement achieved by KCSCD over the strongest baseline. * and ** indicate that the improvements over the strongest baseline are statistically significant, with $p < 0.05$ and $p < 0.01$, respectively.

Model	ASSIST17			ASSIST09			Junyi		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
DINA	0.6340	0.4966	0.7293	0.6905	0.4822	0.7247	0.6568	0.4591	0.7567
IRT	0.6886	0.4633	0.7396	0.7011	0.4756	0.6897	0.7609	0.4176	0.7997
MIRT	0.7110	0.4458	0.7750	0.7227	0.4521	0.7519	0.7649	0.4072	0.8083
NCD	0.6689	0.4867	0.7139	0.7205	0.4646	0.7261	0.7390	0.4174	0.7966
RCD	0.7051	0.4418	0.7654	0.7157	0.4316	0.7424	0.7594	0.4084	0.8057
KSCD	0.6640	0.4964	0.7397	0.7162	0.4750	0.7431	0.7463	0.4781	0.7240
Zero-1-to-3	0.6830	0.4574	0.6873	0.7088	0.4532	0.7452	0.7573	0.4051	0.7965
SCDM	0.6664	0.4659	0.7269	0.7287	0.4971	0.7426	0.7480	0.4965	0.7207
ACD	0.7243	0.4271	0.7968	0.7333	0.4258	0.7713	0.7710	0.4007	0.8255
KCSCD	**0.7269****	**0.4254****	**0.7984****	**0.7339****	**0.4209****	**0.7738***	**0.7727***	**0.3975****	**0.8275****
Improve Rate	+0.26%	-0.17%	+0.16%	+0.06%	-0.49%	+0.25%	+0.17%	-0.32%	+0.20%

Table 3
Ablation study of KCSCD on three datasets.

Student state	ASSIST17			ASSIST09			Junyi		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
KCSCD	**0.7269**	**0.4254**	**0.7984**	**0.7339**	**0.4209**	**0.7738**	**0.7727**	**0.3975**	0.8275
w/o deep learning ability	0.7180	0.4313	0.7874	0.7204	0.4308	0.7578	0.7471	0.4149	0.7936
w/o deep knowledge state	0.7178	0.4314	0.7874	0.7311	0.4226	0.7709	0.7721	0.3977	**0.8277**

5.1.4. Evaluation metrics

We compared our model with baseline models using three commonly used evaluation metrics in the cognitive diagnosis domain: Accuracy (ACC), Root Mean Square Error (RMSE), and Area Under the Curve (AUC). These metrics assess the model's prediction accuracy, error, and classification performance, providing a comprehensive evaluation from multiple perspectives. By analyzing these three metrics, we can measure the model's strengths and weaknesses from different aspects, supporting further optimization and improvement.

5.2. Performance comparison (RQ1)

We compare our model with nine baseline models on three datasets, as shown in Table 2. For these baseline models, we adopted the parameter settings recommended in their original papers, reproduced the models accordingly, and reported their experimental results to ensure a fair and rigorous comparison. The results show that our model outperforms existing methods across all datasets and metrics, proving its effectiveness in cognitive diagnosis tasks. KCSCD utilizes the Knowledge Concept Domain Inference Module, the Deep Learning Ability Assessment Module, and the Hybrid Latent Trait Prediction Module to model students' knowledge states and learning abilities, achieving notable improvements in key metrics. This demonstrates the model's superior discriminative and generalization capabilities in predicting students' mastery of knowledge concepts. Additionally, the Hybrid Latent Trait Prediction Module enables KCSCD to integrate knowledge states and learning abilities, ensuring more stable results. These findings validate the model's superiority in cognitive diagnosis and highlight its potential in intelligent education systems, including knowledge mastery prediction, personalized learning, and learning path optimization, all of which enhance teaching quality and learning efficiency.

5.3. Ablation study (RQ2)

In this study, we focus on two pivotal features: students' deep knowledge state and deep learning ability, derived respectively from the proposed "Knowledge Concept Domain Inference Module" and "Deep Learning Ability Assessment Module". To investigate the impact of these features on model performance, we conduct ablation experiments. Table 3 presents the results of the KCSCD model after removing each feature individually. Specifically, we compare model performance when utilizing only one of the two features. The results demonstrate that when either feature is removed, leaving the model to rely exclusively on the remaining feature, overall performance declines to varying degrees. This finding indicates that these two features are not interchangeable but complementary in cognitive diagnosis tasks.

When we removed the deep knowledge state, the model's performance declined. However, on the Junyi dataset, although the ACC slightly decreased, the AUC slightly increased. This suggests that while removing the deep knowledge state feature led to a slight decrease in model accuracy, its impact on the model's ranking ability was relatively small. Especially on the Junyi dataset, the model may rely on other features to perform a more effective ranking. The deep knowledge state provides the model with fine-grained insights into students' mastery of individual knowledge concepts. Without this feature, the model loses the fine-grained ability to assess students' knowledge mastery, leading to a decrease in accuracy.

When the deep learning ability was removed, the model's performance showed a significant decline. Particularly on the Junyi dataset, both ACC and AUC decreased by 3.31% and 4.09%, respectively. This change indicates that after removing the deep learning ability, the model lost its understanding of how students approach problems of varying difficulty. Especially when handling complex tasks, the absence of this feature prevents the model from fully assessing the students' cognitive processes and problem-solving abilities. Therefore, when this feature is removed, the model's ability to evaluate problem-solving strategies is limited, leading to a significant drop in performance.

Ablation experiments further validate the complementarity of deep knowledge state and deep learning ability. In cognitive diagnosis tasks, these two features jointly contribute to a comprehensive assessment of students' learning status. Model accuracy deteriorates when either feature is removed.

5.4. Hyperparameter analysis (RQ3)

We conduct Hyperparameter analysis experiments on the learning rate and knowledge concept embedding dimension of the KCSCD model

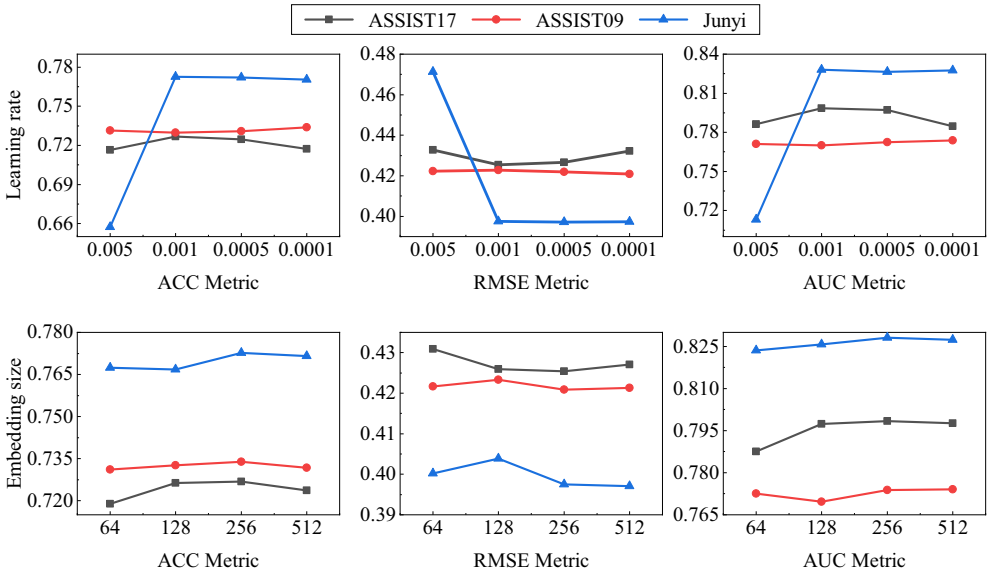


Fig. 3. Performance on different learning rate and embedding size.

across three datasets. The value ranges for the learning rate and knowledge concept embedding dimension are $\{0.005, 0.001, 0.0005, 0.0001\}$ and $\{64, 128, 256, 512\}$, respectively. Fig. 3 illustrates how the output of the KCSCD model changes with different values of these two parameters. From the Fig. 3, we can conclude that when the learning rate is 0.001, the model performs best on all three datasets. Increasing or decreasing this parameter further typically leads to a drop in performance, possibly because a high learning rate causes instability during training, while a smaller rate results in slower convergence, preventing the model from reaching optimal performance. For the study of the knowledge concept embedding dimension, we observe that as the dimension increases from 64 to 256, the model's performance improves, reaching its optimal performance at 256. However, when the dimension further increases to 512, the model's performance declines. This suggests that a higher dimension does not necessarily lead to significant performance improvement and may even negatively impact the results. Through this experiment, we find that setting low dimensions for knowledge concept embeddings may reduce the model's ability to capture data features, leading to performance degradation. Conversely, setting excessively high dimensions may introduce excessive irrelevant information into the embeddings, which also causes performance decline. Overall, the model is relatively sensitive to changes in the knowledge concept embedding dimension.

5.5. Performance on the knowledge concept cold-start problem (RQ4)

To evaluate KCSCD's performance in a knowledge concept cold-start scenario, we preprocess the original dataset to mimic a situation where students have no prior interaction with certain knowledge concepts. Specifically, we exclude interaction records of knowledge concepts practiced by students in the training data from the test data. This results in the test set containing only MKC, aligning the experimental setup with the cold-start scenario assumptions. To evaluate KCSCD's performance in this scenario, we compare it with two cognitive diagnosis models, ACD and RCD, which performed well in the first experiment. Experiments were conducted on three real-world educational datasets using the preprocessed data, and the results are summarized in Table 4.

Experimental results demonstrate that the KCSCD model outperforms both ACD and RCD models on the ASSIST09 and Junyi datasets. On the ASSIST17 dataset, although the RMSE metric of the KCSCD model is slightly lower than that of the ACD model, its ACC and AUC metrics still maintain superiority. These findings demonstrate that the KCSCD model effectively captures intrinsic relationships among knowledge concepts by constructing a similarity scoring matrix. This design aligns with the objective pattern that students often exhibit similar cognitive behaviors and ability levels toward conceptually related knowledge. As a result, even in the absence of student-exercise interaction logs, KCSCD can effectively infer students' mastery of MKC and make accurate predictions for related exercises by leveraging the relationships between knowledge concepts and students' performance on interaction knowledge concepts (IKC). This not only validates the effectiveness of KCSCD in the knowledge concept cold-start scenario but also highlights its robustness in handling data sparsity and cold-start issues, providing a more adaptable solution for cognitive diagnosis tasks.

5.6. Representation visualization (RQ5)

We design a visualization experiment to visually analyze the model's decision-making process, performing analysis from two dimensions encompassing the intrinsic connections among knowledge concepts and individual students' decision-making outcomes. In terms of interconnections between knowledge concepts, based on the ASSIST09 dataset, we randomly selected 10 knowledge concept samples from the similarity matrix obtained after training. A blue-white-red dual-color gradient

Table 4
Performance of RCD, ACD and KCSCD in the knowledge concept cold-start scenario.

Model	ASSIST17			ASSIST09			Junyi		
	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC
RCD	0.6505	0.4787	0.7123	0.7465	0.4034	0.7536	0.7594	0.4084	0.8057
ACD	0.6704	**0.4603**	0.7383	0.7906	0.3881	0.7846	0.7710	0.4007	0.8255
KCSCD	**0.6715**	0.4605	**0.7445**	**0.8006**	**0.3770**	**0.8017**	**0.7727**	**0.3975**	**0.8275**

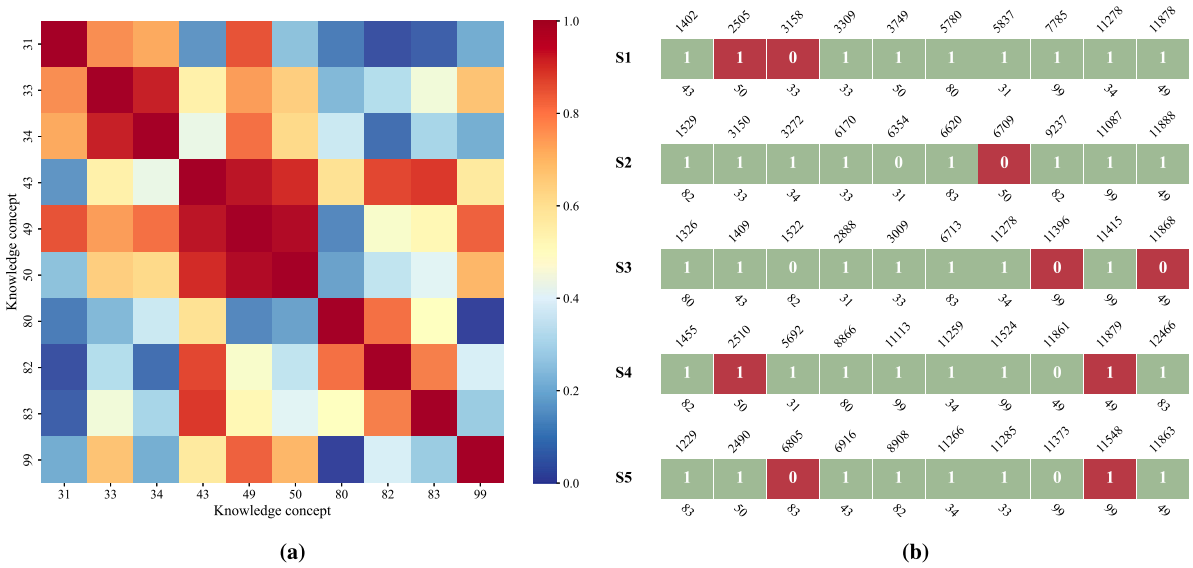


Fig. 4. (a) Similarity heatmap between sampling concepts in the ASSIST09 dataset, similarity scores range from 0 to 1, with higher values indicating stronger similarity between concepts. (b) Sampled student response records and model prediction results in the ASSIST09 dataset. The 0s and 1s in the grid represent students' actual responses, with red indicating model prediction errors and green indicating correct predictions. The numbers above and below each cell denote the exercise ID and knowledge concept ID, respectively.

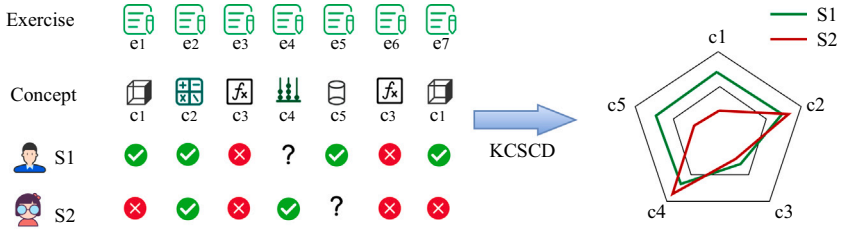


Fig. 5. The diagnosis results of KCSCD.

mapping scheme was employed to quantitatively represent the similarity between concepts, with the visualization results shown in Fig. 4(a). The dark blue region corresponds to weak similarity between concepts, while the red region indicates strong similarity. At the level of individual student decision analysis, as shown in Fig. 4(b), 10 exercise interaction records from 5 randomly selected students in the ASSIST09 test set were visualized alongside the model's predictions for these records. All knowledge concepts associated with the exercises are included within the visualization scope of Fig. 4(a). Red markers indicate incorrect model predictions, while non-red markers denote correct predictions. The values "1" and "0" correspond to the student's correct or incorrect response to the exercise, respectively. Each block displays the exercise ID above and the associated knowledge concept ID below.

Results indicate that knowledge concepts 43, 49, and 50 exhibit significant block-level similarity, confirming the model has clustered these three concepts into a strong cognitive association cluster. If a student demonstrates strong mastery on a problem corresponding to any concept within this cluster, the model can infer that the student also possesses strong knowledge mastery of other strongly associated concepts within the cluster. For example, student $S5$ demonstrated mastery on exercises corresponding to both knowledge concepts 43 and 50. Based on the intrinsic logic of the strong cognitive association cluster, the model successfully predicted that this student also possessed mastery of exercises related to concept 49 within the same cluster, with the prediction matching actual performance. For student $S4$, the model predicted incorrect responses to related exercises based on their unmastered status for concept 49 and its cluster concepts. However, in practice, student $S4$ correctly answered exercise 2510 (associated with concept 50) and exercise 11,879 (associated with concept 49). This discrepancy arises because different exercises corresponding to the same knowledge concept may vary in difficulty, leading to cognitive bias in the model.

The above visualization results and analytical conclusions not only help users gain deeper insight into the internal logic of the model's prediction mechanism but also provide data-driven empirical support and a clear direction for subsequent iterative optimization of the model.

5.7. Case study (RQ5)

In this section, we present a diagnostic example of KCSCD on the ASSIST17 dataset. This case study aims to explore how KCSCD predicts students' mastery of interaction missing knowledge concepts. Fig. 5 presents a diagnostic example of the KCSCD model, which includes response logs from students $S1$ and $S2$ in the ASSIST17 dataset. In this example, student $S1$ has not interacted with knowledge concept $c4$, while student $S2$ has not interacted with knowledge concept $c5$. Our model's diagnostic results, based on interaction records, indicate that student $S1$ has a high proficiency in knowledge concept $c4$, while student $S2$ has a low proficiency in knowledge concept $c5$. We believe there are two reasons for these results. First, our model constructs a knowledge concept similarity matrix based on student interaction records, identifying a high similarity between knowledge concept $c2$ and $c4$, as well as between knowledge concept $c1$ and $c5$. Additionally, students' performance on exercises containing similar knowledge concepts is expected to be similar. Student $S1$ correctly answered exercise $e2$, which

includes knowledge concept $c2$, while student $S2$ answered exercises $e1$ and $e7$ incorrectly, both of which include knowledge concept $c1$. As a result, the model infers that student $S1$ has a high proficiency in knowledge concept $c4$, while student $S2$ has a low proficiency in knowledge concept $c5$. Secondly, student $S1$ has a higher accuracy rate than student $S2$. As a result, the model considers student $S1$ to have a higher learning ability, which positively influences the prediction. In contrast, student $S2$ has a high error rate in the interaction logs, leading the model to infer that they have not yet mastered knowledge concept $c1$ and that their learning ability is insufficient to correctly respond to exercise $e5$. By diagnosing students' proficiency in interaction missing knowledge concepts, our model demonstrates its effectiveness in addressing the knowledge concept cold-start problem. This predictive capability not only helps educational systems more accurately meet the learning needs of each student but also supports instructional decisions, such as designing personalized learning plans and implementing timely interventions.

6. Conclusion

This paper proposes a Knowledge Concept Cold-Start Cognitive Diagnosis Model (KCSCD) to address a critical issue often overlooked by existing methods: when new knowledge concepts are introduced, students' practice records are typically blank, making it impossible for models to effectively diagnose and predict using existing data. KCSCD comprehensively considers both interaction and interaction missing knowledge concepts. It designs a Knowledge Concept Domain Inference Module to model students' deep knowledge state and a Deep Learning Ability Assessment Module to model students' deep learning ability, with a final weighted fusion through a gating mechanism. Experimental results demonstrate that KCSCD outperforms existing methods across three real-world educational datasets, with significant improvements, especially in terms of interpretability.

CRediT authorship contribution statement

Miao Zhang: Formal analysis, Resources, Writing - review and editing. **Huihuan Li:** Validation, Writing - original draft. **Lele Zheng:** Data curation. **Shunfeng Tan:** Writing - review and editing. **Chao Yang:** Supervision. **Kui Xiao:** Funding acquisition, Methodology, Supervision. **Zhifang Huang:** Investigation. **Zhifei Li:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported in part by the National Natural Science Foundation of China (Grant No.62377009, 62407013), the Key Laboratory of Intelligent Sensing System and Security (No. KLISSS202504), and the Hubei Provincial Key Laboratory of Artificial Intelligence and Smart Learning (No. 2025AISL001).

Data availability

Data will be made available on request.

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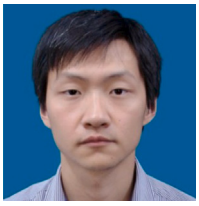
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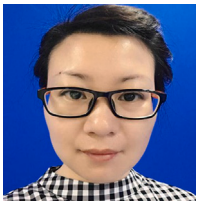
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